

Toward a Theory of Groupoid Graded Rings

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Abstract. We discuss initial steps toward the development of the theory of object-unital groupoid-graded rings. Although such rings do not, in general, possess a global identity element, they admit an appropriate system of local units and we define local chain conditions, instead of global ones.

Among our main contributions, we present a graded version of the Wedderburn–Artin Theorem. In addition, we develop the theory of the graded Jacobson radical, establishing results on different types of nilpotency conditions and relating them with the local chain conditions and proving graded analogues of the Hopkins–Levitzki Theorem about the equivalence of the local chain conditions on graded rings and graded modules.

Despite notable similarities with the classical theory of rings, the groupoid-graded setting exhibits significant differences, stemming largely from the fact that our rings are typically infinite direct sums of right ideals.

This is ongoing joint work with Zaqueu Cristiano from the University of São Paulo and Wellington Marques de Souza from the Federal University of Mato Grosso do Sul.

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Let R be a unital associative ring with Jacobson radical $J(R)$. Important results in Classical Ring Theory are the following, see for example [4].

Teorema 1 (Wedderburn–Artin Theorem). *R is a simple right artinian ring if and only if $R \cong M_n(D)$ where D is a division ring and $n \in \mathbb{Z}_{>0}$.* □

Teorema 2. *R is a semisimple ring if and only if R is a right artinian and $J(R) = 0$.* □

Teorema 3. *If R is a right artinian ring, then $J(R)$ is a nilpotent ideal of R and $R/J(R)$ is a semisimple ring.* □

Teorema 4 (Hopkins–Levitzki Theorem). *Suppose that $R/J(R)$ is a semisimple ring and $J(R)$ is a nilpotent ideal of R . Then the following statements are equivalent for a right R -module M_R :*

- (1) M_R is artinian.
- (2) M_R is noetherian.

In particular, if R is a right artinian ring, then R is a right noetherian ring. □

We generalize those theorems to the groupoid graded context where rings are assumed to be associative but not necessarily unital. We proceed to define the objects that we will be considering and the main obstacles to obtain our results.

A *groupoid* Γ is a small category in which every morphism is an isomorphism. We will denote the set of objects of Γ by Γ_0 and we identify them with its identity morphisms. Thus, the identity morphism of $e \in \Gamma_0$ will be denoted again by e . If $e, f \in \Gamma_0$ and γ is a morphism from e to f , we will write $d(\gamma) = e$.

We say that the ring R is a Γ -graded ring if there is a family $\{R_\gamma\}_{\gamma \in \Gamma}$ of additive subgroups of R such that $R = \bigoplus_{\gamma \in \Gamma} R_\gamma$ and $R_\gamma R_\delta \subseteq R_{\gamma\delta}$, for each $\gamma, \delta \in \Gamma$. Following [1], we say that R is *object unital* if, for all $e \in \Gamma_0$, the ring R_e is unital with identity element 1_e , and for all $\gamma \in \Gamma$ and $r \in R_\gamma$, we have $1_{r(\gamma)}r = r1_{d(\gamma)} = r$. It follows from [5, Proposition 2.1.1] that R is unital if and only if the set $\{e \in \Gamma_0 : 1_e \neq 0\}$ is finite. Observe that R has the following decomposition as a direct sum of right ideals

$$R = \bigoplus_{\gamma \in \Gamma} R_\gamma = \bigoplus_{e \in \Gamma_0} \underbrace{\left(\bigoplus_{r(\gamma)=e} R_\gamma \right)}_{R(e)} = \bigoplus_{e \in \Gamma_0} R(e).$$

We say that R is right Γ_0 -artinian if $R(e)$ satisfies the descending chain condition for its graded submodules for each $e \in \Gamma_0$. Similarly, one can define when R is *right Γ_0 -noetherian*. The *graded Jacobson radical* of R is the intersection of all maximal graded right ideals of R . One can show that it is in fact an ideal.

We are able to show a Wedderburn-Artin Theorem for R when it is *gr-semisimple*, i.e. R is a (possibly infinite) sum of gr-simple R -modules.

Unlike in the classical case, the graded Jacobson radical of a Γ_0 -artinian ring is not necessarily nilpotent. However, we present a Hopkins-Levitzki type theorem for *gr-semiprimary rings*, that is, graded rings such that $R/J(R)$ is gr-semisimple and $J(R)$ is *objectwise nilpotent* in the following sense: for each $e \in \Gamma_0$, there exists a natural number n_e such that $1_e J^{n_e} = J^{n_e} 1_e = 0$. We observe that a right Γ_0 -artinian ring does not imply being right Γ_0 -noetherian or having an objectwise nilpotent graded Jacobson radical. We show that a ring that is both right and left Γ_0 -artinian is always gr-semiprimary.

This is an ongoing project with Zaqueu Cristiano from the University of Sao Paulo and Wellington Marques de Souza from the Federal University of Mato Grosso do Sul. Some of our results can be found in [2, 3].

References

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